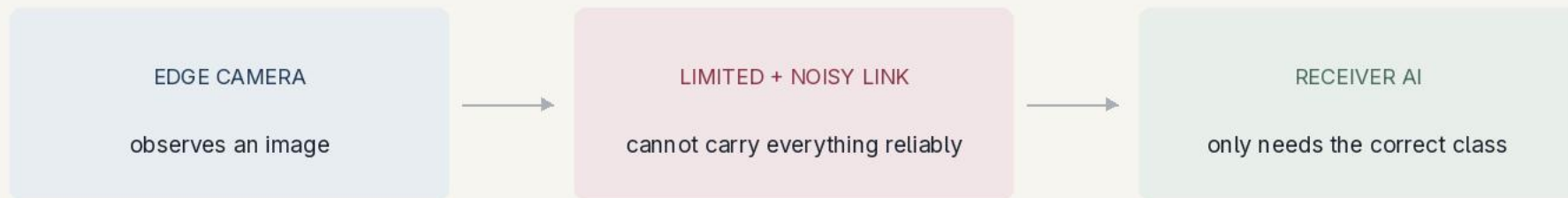


# Semantic Communication for Edge AI

Can a machine communicate what matters for a task—without sending every pixel perfectly?



## THESIS

Train the sender and receiver together so the information needed for classification survives the link—then compare it fairly with a strong conventional system.

A simulation-first capstone with an optional real-radio demonstration later.

# The AI is only as useful as the link feeding it

Imagine a remote camera sending images to a server that decides what it sees.

## REAL-WORLD TENSION

- 1** The sensor is constrained      Bandwidth, latency and energy are limited at the edge.
- 2** The wireless link is imperfect      Noise can corrupt a short message or make decoding fail.
- 3** The receiver has a narrow goal      For classification, the final question is often simply: what is in the image?

## DESIGN QUESTION

Why spend the entire link budget reconstructing every pixel if the receiver ultimately needs a task decision?

The project tests this question; it does not assume the learned method must win.

Goal: preserve classification performance when communication resources are scarce and the link becomes noisy.

# From bit-perfect delivery to task success

CONVENTIONAL COMMUNICATION   Compress the image → protect the bits → reconstruct the file → classify it

Primary objective: recover source bits and pixels

TASK-ORIENTED COMMUNICATION   Learn a representation → send it through noise → recover what the task needs

Primary objective: retain information useful for classification

## EXPECTED SIGNATURE

As the link worsens, a separated chain may hit a decoding cliff; a learned joint system may degrade more gradually.

## IMPORTANT CONTROL

A digital system can also send learned features—so the experiment includes that third arm.

Research question: does the task-oriented joint design help after bandwidth, noise and comparison strength are matched fairly?

# How prior research leads to this project

## Finite blocklength

[1–4]

Rate backs off from capacity; source and channel dispersions interact.

OPEN ISSUE

Need measured practical short-packet loss.

## Learned compression

[11–17]

End-to-end transforms and entropy priors improve rate–distortion.

OPEN ISSUE

Reconstruction optimization is not task optimization.

## DeepJSCC

[5–10, 24]

Continuous learned mappings exhibit graceful degradation under mismatch.

OPEN ISSUE

Most image studies foreground reconstruction quality.

## Task-oriented communication

[18–23]

Learned representations can preserve remote inference under link constraints.

OPEN ISSUE

Task awareness must be separated from joint coding.

Gap carried forward: three-way attribution at equal channel uses, with a strong adaptive classical comparator and paired failures retained.

# What we will build—and what counts as success

## OBJECTIVES

1. Build a reproducible normalized-AWGN DJSCC pipeline.
2. Implement a standards-derived, non-strawman JPEG 2000 + NR LDPC baseline.
3. Tune classical quality, code rate and modulation on validation only.
4. Add the task-aware digital feature control for attribution.
5. Evaluate paired image outcomes once on the sealed test split.
6. Report positive, null or negative outcomes under one frozen protocol.

## H1 · PRIMARY

### Low-SNR separation: learned – classical\_adaptive

A point qualifies when  $\sqrt{N} \cdot \Delta(s) / \sigma(s) > 1.96$ . H1 is supported only if the longest run  $R_{\text{obs}}$  contains at least three consecutive qualifying points at or below the training SNR and the calibrated run p-value  $\leq 0.05$ .

Effect size of record: mean paired accuracy difference over the full low-SNR region.

#### H2 Graceful vs cliff

Fixed classical curve drops  $\geq 30$  pp while learned drops  $\leq 15$  pp over the frozen window.

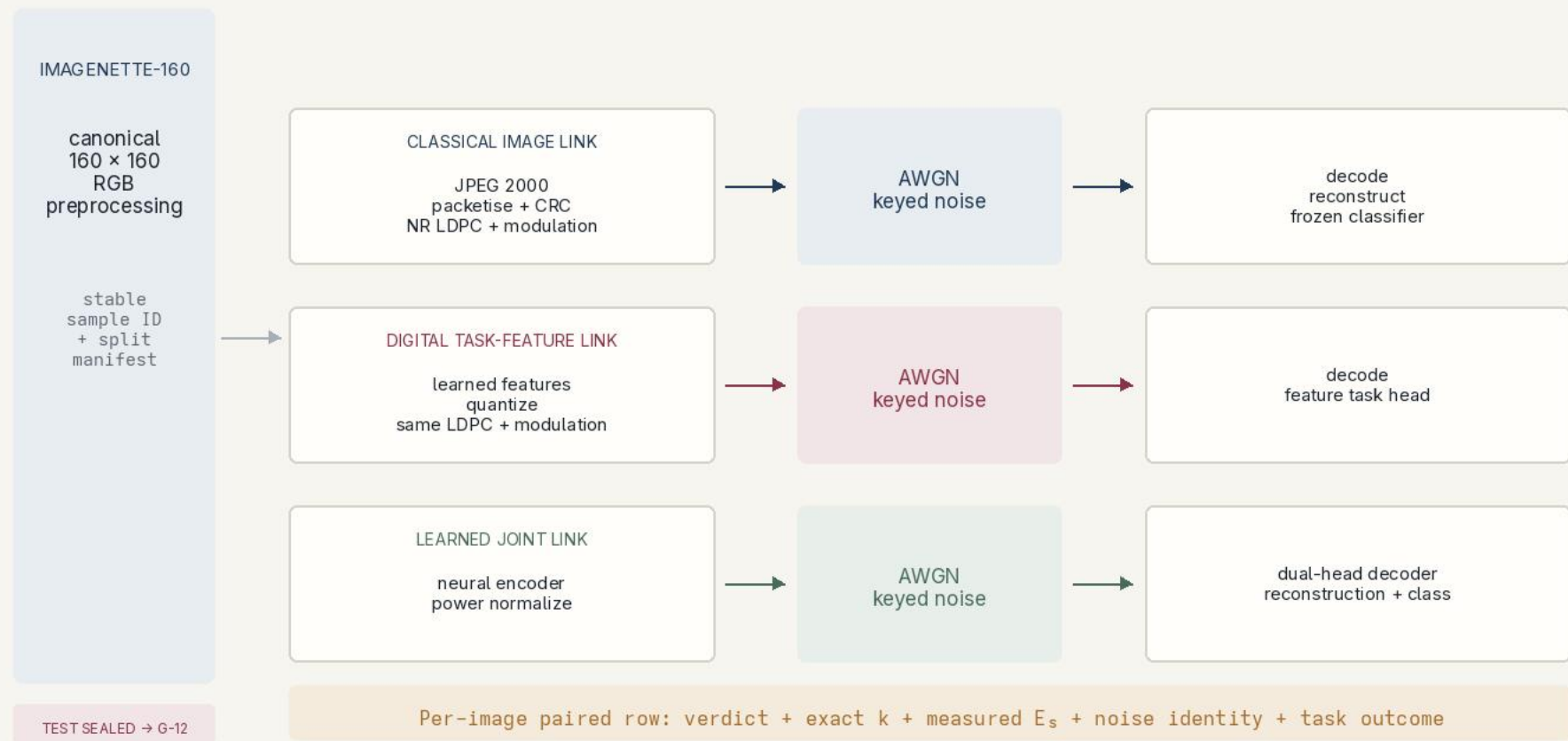
#### H3 Convergence

Paired gap trends toward zero across SNR; crossover is reported, not required.

#### H4 Attribution

Credit joint coding only if DJSCC also exceeds the task-aware digital control under the H1 rule.

# The experiment in one picture





# How we keep the comparison fair

## Equal resource

Same complex-symbol budget  $k$  and  $E_s/N_0$  definition; realized symbol energy is logged.

## Strong comparator

JPEG 2000 quality, LDPC rate and BPSK/QPSK/16-QAM tune per SNR on validation.

## Exact accounting

Container, CRC, code-block and filler bytes count; decode failures remain in the denominator.

## Measured BLER

Every required physical identity is characterized; missing evidence is never treated as zero BLER.

## Outage policy

Constant-class fallback is measured from validation counts—100/1,000—not assumed as 1/10.

## Classifier control

Clean and artifact-finetuned classifier passes separate codec shift from link failure.

## Attribution arm

Quantized learned features traverse the same digital chain, isolating task awareness from joint coding.

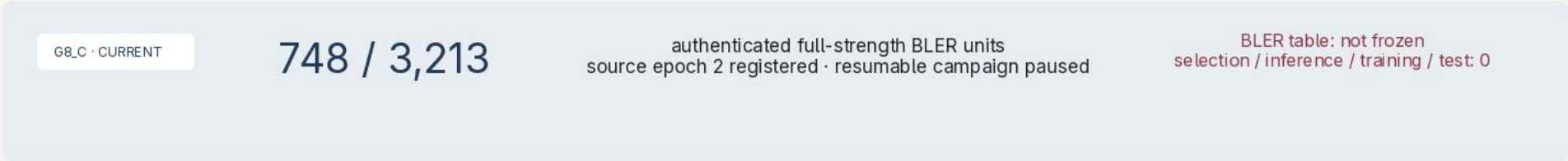
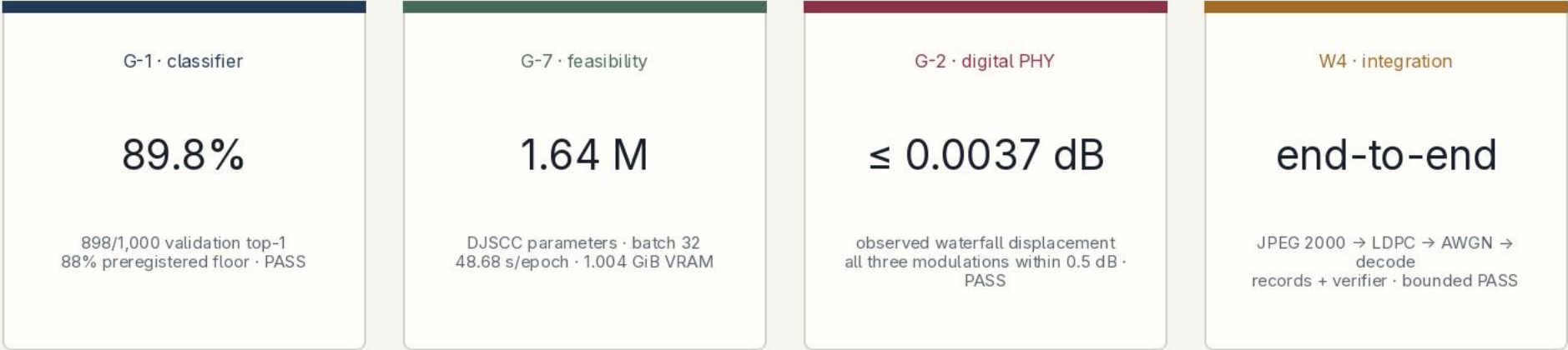
## One test campaign

All choices freeze before G-12; paired inference retains complete image × system trajectories.

Fail-closed rule: uncharacterized, infeasible and failed cases are explicit outcomes—not silently dropped rows.

# Engineering and gate evidence—not learned-vs-classical results

NO HEADLINE COMPARISON EXISTS YET · bounded smoke is integration evidence, not scientific outcome evidence



Interpretation: the reference task, computational budget and digital chain are verified; the comparison itself remains ahead.



# Gate-ordered path from characterization to final report

Later work is shown as planned—not backfilled as complete for Review 1.



Critical path: G8\_C → G-8 → training →  $\lambda$  calibration → ER-9/G-11 → validation rehearsal → G-12 → one test campaign → G-5 → report

W16 is report contingency—not room for new experiment scope.

# What this First Review should confirm

## RESEARCH SCOPE

### Project boundary

One narrow task: classify an image after communication over a bandwidth-limited noisy link.

The sender runs an encoder; the receiver owns the task head. That deployment split is fixed.

Tier 1 uses a normalized simulated AWGN channel so the scientific comparison is controlled and reproducible.

The result may be positive, null or negative.

## IMPLEMENTATION

### Deployment ladder

TIER 1 · required  
Simulation-first, offline and reproducible.

TIER 2 · stretch after G-5  
Conducted SDR I/Q replay; prerecorded outcome expected.

TIER 3 · stretch  
Edge placement only if latency and procurement gates pass.

Candidate: HackRF One + attenuation chain + RTL-SDR; no purchase yet.

## PANEL ASK

### Decision requested

Confirm that the project asks the right question and compares the systems fairly:

- image-classification is the task
- communication resources are matched
- the classical baseline is strong
- the digital feature arm supports attribution
- simulation-first Tier 1 is sufficient

Guide acknowledgement: PENDING

## TECHNICAL BOUNDARY

OpenJPEG 2.5.4 + TS 38.212-derived LDPC/rate matching over abstract AWGN—not a full 5G NR link claim

No guaranteed win, measured energy saving or real-radio generalization; completion means executing the preregistered protocol.

# Explicit First Review rubric-to-slide map

| CRITERION         | SLIDES     | EVIDENCE EXPOSED IN THE DECK                                       |
|-------------------|------------|--------------------------------------------------------------------|
| Motivation        | 2–3        | Short packets, fixed $k$ , task accuracy, finite-blocklength costs |
| Objectives        | 3, 5–7, 10 | Three-arm build, fair protocol, attribution control, decision ask  |
| Hypothesis        | 5          | Exact H1 run rule; H2–H4 preregistered; crossing not required      |
| Problem Survey    | 2, 4, 12   | 30-source synthesis across four literature families                |
| Subject Knowledge | 3–8, 10    | AWGN, LDPC, JPEG 2000, fairness, evidence and scope boundary       |
| Time Plan         | 9          | Gate-ordered critical path and fixed review/report dates           |

Final PASS still requires the real PPT review, four-member rehearsal, guide acknowledgement and annotated review-1-basis tag.

# Selected references and evidence provenance

SELECTED LITERATURE

[1] Shannon, A Mathematical Theory of Communication, 1948.

[2] Gastpar, Rimoldi & Vetterli, To Code, or Not to Code, 2003.

[3] Polyanskiy, Poor & Verdú, Finite Blocklength, 2010.

[4] Kostina & Verdú, Lossy JSCC at Finite Blocklength, 2013.

[5] Boursoulatz, Kurka & Gündüz, DeepJSCC, 2019.

[6] Kurka & Gündüz, DeepJSCC-f, 2020.

[7] Kurka & Gündüz, Bandwidth-Agile DeepJSCC, 2021.

[11] Ballé, Laparra & Simoncelli, Learned Compression, 2017.

[12] Ballé et al., Scale Hyperprior, 2018.

[17] Blau & Michaeli, Perception-Distortion, 2018.

[18] Jankowski, Gündüz & Mikołajczyk, Wireless Retrieval, 2021.

[19] Shao, Mao & Zhang, Task-Oriented Edge Inference, 2021.

[20] Xie et al., Deep Learning Enabled Semantic Communication, 2021.

REPOSITORY EVIDENCE

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| G-1    | results/reference_classifier/g1_adjudication.json<br>89.8% validation; test sealed  |
| G-7    | results/profiling/g7_djscc_profile.json<br>1.64M params; measured epoch/VRAM        |
| G-2    | results/baseline/g2/g2_adjudication.json<br>golden vectors + BLER conformance       |
| W4     | results/baseline/w4/integration_adjudication.json<br>bounded end-to-end integration |
| G8_C   | results/baseline/g8/campaign_state.json<br>authenticated characterization cursor    |
| Spec   | spec/SPEC.md<br>normative hypotheses and requirements                               |
| Plan   | docs/gantt-plan.md<br>dates, gates and current status                               |
| Deploy | docs/deployment-dossier.md<br>simulation-first + SDR stretch                        |

Refresh G8\_C status from instructions/RESUME.md on submission day.